The Hydra Heads

AI-Powered Recovery Intelligence

Camera Pose Detection · Claude AI Protocols · Smart Thermal Device Control

◎ ASSESS

Real-time ROM & asymmetry via camera

✦ PRESCRIBE

Claude AI generates personalized protocols

⊕ TREAT

MQTT controls dual-pad device

Team: The Hydra Heads

React Native · Expo · Claude AI · MQTT · rPPG · MoveNet Pose

A $47B Industry Running on **Guesswork & Gut Instinct**

Physical therapy & recovery clinics face a critical data gap — no objective biomechanical baseline exists for device personalization.

Subjective Intake

Practitioners rely on 1-10 pain scales and verbal self-reports. No objective biomechanical baseline.

Slow Protocols

Average intake + ROM assessment takes 15-20 minutes. Manual goniometry is tedious and inconsistent.

Dumb Devices

Thermal/LED recovery devices operate on fixed factory presets — zero personalization per patient.

$47B	68%	15 min	$1.2B
Physical Therapy Market	Report inconsistent outcomes	Avg manual ROM assessment	Red light & thermal device market

THE CORE BOTTLENECK

Recovery devices prescribe generic, one-size-fits-all therapy to bodies that are uniquely asymmetric and injured. There is no closed loop between patient biomechanical data and device output.

Close the Loop:
Camera → AI → Device

◎ CAMERA
ASSESSMENT
MoveNet 17-pt pose + contactless rPPG heart rate and breath rate

→

✦ ROM +
ASYMMETRY
Pose Assessment: Shoulder/knee angles, side-to-side delta

→

CLAUDE AI
PROTOCOL
Patient data + camera → JSON protocol

→

⊕ MQTT DEVICE
CONTROL
Sun/Moon pad, intensity & cycle config

☀ Practitioner Mobile App (React Native / Expo)

- Live camera + skeleton pose overlay + speech to text translation (ElevenLabs)
- Shoulder & knee ROM angle badges (live)
- rPPG heart rate from facial pixels (works on all skin tones)
- 60-second AI patient intake + report of session
- Session control: Start / Pause / Stop

☾ Hydrawav3 Dual-Pad Device

- Sun Pad — heating + red LED 660nm
- Moon Pad — cooling + blue LED 450nm
- MQTT topic: HydraWav3Pro/config
- PWM control: gentle / moderate / intense
- 9-min sessions, 3-cycle orchestration

CLAUDE AI CORE

Prompt: patient demographics + camera ROM + asymmetry flags → returns JSON: pad placement, intensity goal, coaching tip & retest focus — in under 2 seconds.

Built to Scale with the Hydrawav3 Ecosystem

REVENUE MODEL · SaaS + Hardware Partnership

FREE
Patient App

- ✓ Daily AI coaching tips
- ✓ Session history & tracking
- ✓ Self check-in with ROM guide
- ✓ Between-visit motivation

Patient retention driver

RECOMMENDED

$49/mo
Practitioner

- ✓ AI protocol generation
- ✓ Camera ROM assessment
- ✓ Device MQTT control
- ✓ Outcome reporting + analytics

Per provider — core revenue

$199/mo
Clinic / Enterprise

- ✓ All practitioner features
- ✓ Multi-provider management
- ✓ Priority Claude API access
- ✓ White-label branding

Per location — MedSpa ideal

CLIENT ROI — AT $49/MO SUBSCRIPTION

Time saved per intake	15 min → 60 sec	Protocol generation	10 min manual → <2 sec AI
Extra patients per day	+2–3 additional sessions	Monthly revenue uplift (est.)	$450–$900 vs $49 subscription

TAM	SAM	SOM
$47B	$4.7B	$47M
US PT + wellness market	Tech-enabled recovery clinics	First 3yr with device deals

First 1,000 Customers:
The Device Distribution Play

TARGET SEGMENTS

Physical Therapists
HIGH

7,500+ outpatient PT clinics. Use as a differentiated tech tool for existing patients.

Chiropractors
HIGH

70,000+ chiropractors. Wellness-language protocols fit chiro legal compliance.

Sports Trainers
MED

Team/individual trainers seeking objective athlete ROM tracking and recovery tools.

MedSpa / Wellness
MED

Fastest growing segment. Thermal + LED is a premium add-on. High willingness to pay.

ACQUISITION ROADMAP · PATH TO 1,000 CUSTOMERS

Phase 1 · Month 1–3

Device Partnership Seeding

Partner with 10 Hydrawav3 resellers. Each has 100+ clinic customers. Warm intro email campaign.

Customers: 0 → 100

Phase 2 · Month 3–6

Community & Conference

Exhibit at APTA, NSCA, regional wellness expos. 60-day free trial. Collect outcome case studies.

Customers: 100 → 400

Phase 3 · Month 6–12

Referral Engine

Freemium patient app drives practitioner referrals. Patients follow their providers onto the platform.

Customers: 400 → 1,000

DEFENSIBLE MOAT

Outcome data accumulates per patient → AI protocols improve → device manufacturers embed us → network effect locks in practitioners.

Competing For:
AI + Wellness / Hydrawav3 Hardware Track

SPONSOR TRACK: Hydrawav3 Hardware Challenge

Build the most innovative application for the Hydrawav3 dual-pad device using the provided MQTT API.

Our answer: Close the AI loop between patient body data and device output via pose detection + Claude protocol generation.

PROMPT REQUIREMENT → SOLUTION MATCH

✓ Use the Hydrawav3 MQTT API	deviceLogin → JWT auth → publish to HydraWav3Pro/config with full PWM + vibration payload.
✓ Show device control in action	SessionScreen sends buildSessionConfig() → device Start/Pause/Stop via playCmd 1/2/3/4 over MQTT.
✓ Meaningful AI layer	claude-sonnet-4 generates pad placement + intensity + goal from camera ROM in under 2 seconds.
✓ Real-world use case	Camera assessment → intake → AI protocol → device session → outcome tracking → daily coaching.
✓ Original & creative	rPPG heart rate from camera face pixels + pose asymmetry = a fully novel biometric intake pipeline.

WHY THE HYDRA HEADS WIN

Computer Vision First

No wearables. Camera is the only sensor — zero friction beyond the recovery device itself.

Claude at the Core

AI isn't decorative — it makes the clinical decision: which pad goes where, at what intensity.

Full Stack Demo

Camera → AI → MQTT → Device → Patient App. Every layer of the prompt is shipped and demoed.